Weather Detection Module & Flowering Guidance for Anthurium Farming

Project Id: 25-26J-357

Project Proposal Report

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DECLARATION

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We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Soysa H. R. T.	IT22109958	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

....

Signature of the supervisor: Date: 29/8/25

ABSTRACT

This study describes an Intelligent Decision Support System (IDSS) to promote the sustainable farming of the ornamental plant Anthurium with a Weather Detection and Flowering Guidance module. Anthurium production systems depend heavily on environmental conventions since Anthurium is highly affected by changes in environmental conditions. Because of its sensitivity to climate, Anthurium farming groups require time-sensitive information, flexibility, and coordination to respond to changing environmental situations. Conventional Anthurium farming relies heavily on a farmer's instinct; they plan their cultivating cycle based on the weather, using historical references before farming decisions are made. As weather becomes more unpredictable, conventional practices lead to inefficient practices.

The Weather Detection module collects and handles data on environmental conditions in real-time, including temperature, humidity, rainfall, sunlight hours, and wind speed, from trusted Weather Forecast APIs. The Weather Detection module applies data preprocessing techniques, such as cleaning, and standardization and other formatting methodologies to improve data quality and accuracy. Finally, the real-time data is assessed through forecasting and classification models, and utilized to examine both short-term and long-term changing environmental conditions.

In addition, the Flowering Guidance module utilizes predictive modeling and time-series forecasting to estimate flowering frequency and spike counts. Environmental conditions are classified as good, moderate, or poor, this allows the system to provide growers with timely insight. Recommendations regarding irrigation, shading, and ventilation, are included to reinforce the cultivation decision-making process.

The module is built as a web-based interface in a front-end and a backend API (Flask/FastAPI) which allows growers to view simple dashboards of flowering probability, predictions and cultivation recommendations. Combining AI, Data Science and Smart Agriculture, the Flowering Guidance module offers productivity, resilience and sustainability in Anthurium cultivation.

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LIST OF ABBREVIATIONS

DDS - Decision Support Systems

IoT - Internet of Things

AI – Artificial Intelligence

API - Application Programming Interface

1. INTRODUCTION

1.1 Background & Literature survey

Background

An ornamental crop that is highly cultivated for its market and aesthetic demand is Anthurium. Like other crops, the flowering of the Anthurium and its sensitivity to environmental factors such as temperature, humidity, rainfall, sunlight, and wind speed is noted. Delays in flowering, lower spike quality, and stress to the crop caused by slight fluctuations in these factors result in losses to the growers. Agriculture, as in other crops, is sensitive to an array of climatic factors. Similarly, the growth and yield of crops are proportionally affected by variability in climate. The demand and aesthetic value for which anthurium is highly cultivated is evident as in other ornamental crops. Like other crops, the growth and flowering of anthurium, as well as the subsequent spike quality, is highly sensitive to changes in temperature, rainfall, humidity, sunlight, and even wind speed. A delay in flowering and lower quality spikes is minimally caused by all of these factors, while stress to the crop results in increased losses to the farmers.

Decisions regarding cultivation and flowering time are often taken by manual Vietnamese anthurium farmers based on their own observations and experiences. Such methods are known to work out well under smaller scale conditions and with lesser demands. However, such methods are considered dated and are bound to fail in the long run due to the modern-day challenges of climatic fluctuation, increased demands, and the need for sustainability. Additionally, trial and error are bases for such methods, leading to unnecessary expenditure and wastage of valuable resources like water and electricity.

Literature Survey

The integration of smart technologies into agriculture has been extensively researched, especially in the fields of weather tracking, yield estimation, and decision support. Weather is hazardous to farming and many studies have tried to use accurate weather predictions to enhance crop farming in general. For example, for the purpose of irrigation

scheduling, real-time weather data capture and analysis systems have been employed. As an advanced approach to utilize external climatic data, such systems enable the mitigation of climatic stress in a variety of crops and support the prediction of rainfall and other weather events.

For the purpose of estimating growth stages and flowering behavior, the use of machine learning and data-driven methodologies in crop prediction is gaining attention in recent studies. In the case of floriculture, attention has been given to the cultivation of orchids, chrysanthemums, and Anthuriums in relation to environmental factors such as temperature, humidity, and photoperiod. Predictive models, such as regression analysis and time series forecasting have been used to estimate flowering frequency, yield, and spike development. Such models display the need of understanding the complex relations of plants with the environment and provide methods in enhancing both quantity and quality.

DSS have also gained significant attention as tools for smart agriculture. These systems typically combine predictive algorithms with rule-based logic to generate practical recommendations for farmers. Applications of DSS range from greenhouse climate control to pest management and irrigation scheduling. By integrating environmental data with computational intelligence, DSS platforms have proven effective in providing user-friendly guidance, often through mobile or web interfaces. They serve as a bridge between advanced analytics and the day-to-day practices of farming, ensuring that technological innovations are accessible and actionable.

Although there has been much progress, there are many areas where intelligent systems for ornamental crops have not been applied. Most research has been focused on food crops such as rice, maize, and wheat and has not brought intelligent systems to the same attention for their behalf with ornamental plants. For example, Anthurium is a crop highly sensitive to external environments and would require specific solutions for flowering management and yield increases. Furthermore, many of the intelligent farming systems that exist mostly rely on IoT-based sensors, which can be expensive to deploy and difficult for small and

medium-sized farmers to deploy. This highlights the gap and need for affordable, API supported, and AI-based decision support systems that could be directed towards ornamental crops.

In summary, previous works indicate the strong potential of weather-based forecasting, predictive analytics, and decision support platforms in agriculture in general, while understanding ornamental crop challenges such as those in Anthurium, are still in early exploration. This project intends to close the gap with a Weather Detection and Flowering Guidance module that employs real-time API feeds, predictive modeling, and ultimately, actionable solutions. This has the potential to advance sustainability, increase yield, and put applied and practical data and support in Anthurium growers.

1.2 Research Gap

Despite the advances made in applying Artificial Intelligence and Decision Support Systems to agriculture, much of the research describes the implementation of these systems on staple food crops such as rice, maize, and wheat, rather than ornamental plants. Anthurium, a climate-dependent floricultural crop, has unique flowering patterns and environmental requirements that have not been sufficiently studied in relation to Artificial Intelligence technology. Current intelligent systems in the field of floriculture involve limited scope, greatly dependent on expensive IoT-based sensors, and the eventual implementation of these systems would not be feasible for small- to medium-scale growers.

Additionally, most past research focuses on general descriptions of crop yield prediction, or greenhouse climate management without real-time weather data or flowering guidance systems; the systems that do exist do not provide useful information that merges weather detection with predictive flowering analysis according to those conditions, and even if they were constructed, the current lack of herbaceous production know-how enacted among growers resemble scant information. While some predictive and rule-based models are in existence for particular crops, they fail to produce useful, grower-friendly outputs,

such as a type of irrigation schedule, a shading answer, or type of venting implementation as described in the preceding chapter that would relate directly to flowering optimization. Furthermore, we believe the synergy of API-based weather data and predictive analytics in ornamental horticulture has not been adequately explored. Current reports rarely demonstrate a complete end-to-end approach which connects the collection of data, its preprocessing, its analysis, and the direction towards sensible climatic action via an accessible web-based interface. It is clear that there is a need for an affordable AI and user-friendly solution to provide real-time flowering advice and contribute to sustainable Anthurium agriculture by reducing the gap between intelligent computing and practical agricultural farming.

1.3 Research Problem

Anthurium farming relies heavily on environmental conditions, and flowering timing and frequency have direct effect on the yield, quality, and market value. The traditional farming methods, on which growers utilize mo0re than anything their own experience and eyes, are always inconsistent and often inaccurate given weather (unpredictable and seasonal), reason(s) related to climate change, etc. This inconsistency creates challenges for growers such as a reduction in flowering frequency, reduced spike counts, and therefore crop quality; ultimately resulting, among other drawbacks such as being an inefficient resource, in negative economic returns.

Farming systems that function intelligently have been successfully developed for staple crops, but more strategic studies and applications for ornamental crops such as Anthurium remain limited. Current solutions are misaligned by focusing solely on general yield prediction, or control of greenhouse climates, or expensive setups of IoT-based sensors which are out of reach for small- and medium-sized farmers. Further, while these systems are particularly capable, they mostly offer very little actionable, flowering-specific, assistance. Rather, the current systems integrate weather data in real time with predictive analysis but not as farm and case specific as occurs in conventional ornamental crop

discussion- there is considerable gap between immediate, implementable technology and assessment within technology.

Therefore, the core research problem is the absence of an affordable, intelligent, and data-driven system that can monitor real-time weather conditions, predict flowering patterns, and provide actionable cultivation recommendations to support sustainable Anthurium farming. Addressing this problem will empower growers to make informed, timely decisions that optimize flowering outcomes, enhance yield quality, and promote sustainable agricultural practices in the face of changing environmental conditions.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this research project is to design and implement a Weather-Optimized Cultivation Planner specifically for Anthurium farming. This system is intended to assist farmers in making well-informed decisions by leveraging real-time weather data and understanding the climate-sensitive nature of Anthurium cultivation. By incorporating accurate environmental information, the system aims to enhance the timing and quality of flowering, ultimately improving overall productivity. The approach focuses on bridging the gap between traditional farming practices and modern data-driven solutions, ensuring that farmers can respond effectively to varying climatic conditions and optimize their cultivation strategies.

2.2 Specific Objectives

In order to realize this overarching aim, a few specific objectives are established. First, more insight is aimed to be captured from farmers about their cultivation practices under varying conditions of weather, such as humidity, temperature, and exposures to sunlight, to help identify practical challenges and necessary conditions to facilitate Anthurium production.

The second objective is to connect weather APIs to communicate climate data and information in close to real-time, and trusted and accurate resources are to be provided to farmers to base their decisions upon.

The third goal is to provide notifications and reminders for unacceptable or adverse weather or climate-sensitive farming activities, so that awareness is raised among farmers and informed choices can be made to avoid or pivot to an alternative approach.

The last objective is to provide improved machine learning-based recommendations, as farm- and weather-based real-time feedback and outcomes are continuously analyzed and evaluated. In this way, consistent and improved recommendations are aimed to be offered that are in line with the development of practices and increased productivity over time.

3. METHODOLOGY

3.1 System Architecture

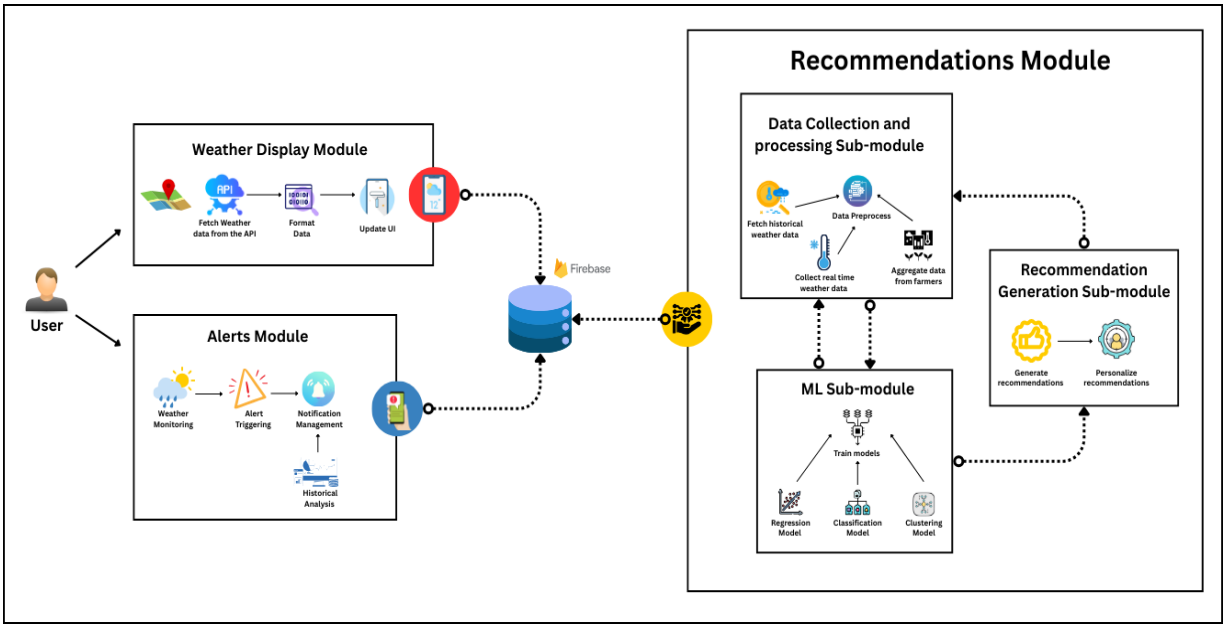

Figure 1 -- System Architecture

3.1.1 Software Solution

Tools and Technologies

- Flutter
- Python
- TensorFlow
- Flask
- Google Colab
- Firebase Firestore
- Weather APIs
- Google Maps SDK

3.1.2 Commercialization

The Weather Detection and Flowering Guidance module of the Intelligent Decision Support System for Sustainable Anthurium Farming can be commercially viable in the AgriTech space. The module uses real-time weather APIs and predictive flowering models to provide growers with actionable data to support cultivation decisions. This solution provides growers focused on Anthurium with an approach to help mitigate unpredictable weather, mitigate yield inconsistency, and deliver flowers meeting market demand for high quality. The solution could be commercialized as a subscription-based mobile/web application that supports farmers, cooperative societies and commercial flower exporters. Future commercialization potential can be explored through partnerships with agricultural advisory services, floriculture, and government agricultural department to further promote the adoption of the system.

Commercialization Strategy:

- Subscription Model: Offer tiered subscription packages (basic, premium) with varying levels of weather insights, flowering predictions, and advisory recommendations.
- Partnerships with Agri-Institutions: Collaborate with agricultural boards, floriculture cooperatives, and government agencies for mass adoption and farmer training.
- Integration with E-Commerce Platforms: Link with online flower marketplaces and exporters to provide farmers with market-driven flowering guidance.
- Freemium Model for Awareness: Provide limited free access to weather alerts to attract farmers, encouraging them to upgrade to premium plans for full guidance.
- Localization: Customize the system for regional weather conditions and local languages to increase usability among rural farmers.
- Data Monetization: Use aggregated weather and crop data insights to support policy makers, researchers, and agri-tech businesses.

3.1.2.1 Future Scope

- Expansion to Other Crops: Extend the model beyond Anthurium to other climate-sensitive ornamental and cash crops.
- AI-Driven Market Forecasting: Incorporate market price prediction and demand forecasting to help farmers align flowering cycles with peak demand.
- IoT Integration: Enhance precision by connecting with affordable IoT devices (e.g., low-cost soil or humidity sensors) in future versions.
- Global Commercialization: Scale the solution to international floriculture markets with customization for different climates.
- Sustainability Monitoring: Integrate modules to measure water usage, carbon footprint, and eco-friendly farming practices.
- Predictive Analytics Dashboard: Offer advanced analytics and visualization tools for large-scale commercial growers and exporters.

4. PROJECT REQUIREMENTS

4.1 Functional requirements

- Weather Data Collection: Collect real-time weather data (temperature, humidity, rainfall, wind speed) via APIs.
- Flowering Prediction: Predict Anthurium flowering patterns based on historical and current weather data.
- Dynamic Guidance Adaptation: Provide actionable recommendations to optimize flowering and spike quality.
- Dashboard & Visualization: Display weather forecasts, flowering predictions, and alerts on a web/mobile dashboard.
- Alert & Notification System: Notify farmers about adverse weather conditions or important cultivation updates.
- Data Management: Allow administrators to update crop and weather-related data efficiently.
- User Feedback System: Collect farmer feedback to improve system guidance and usability.
- APIs and Integrations: Integrate with external services, including weather APIs and market advisory platforms.

4.2 Non-functional requirements

- Performance: Deliver weather updates and flowering predictions quickly and efficiently.
- Usability: The user interface should be simple, intuitive, and accessible to farmers with minimal technical knowledge.
- Scalability: The system should support multiple users simultaneously and accommodate future expansion to other crops.
- Reliability: The system should ensure accurate and consistent data delivery with minimal downtime.
- Security: User data and system data must be securely stored and protected against unauthorized access.
- Availability: The system should be accessible 24/7 with cloud-based hosting.

- Maintainability: The system should be easy to update, including API integration, prediction models, and dashboard features.
- Localization: Support for multiple languages and regional customization to cater to farmers in different areas.

5. GANTT CHART

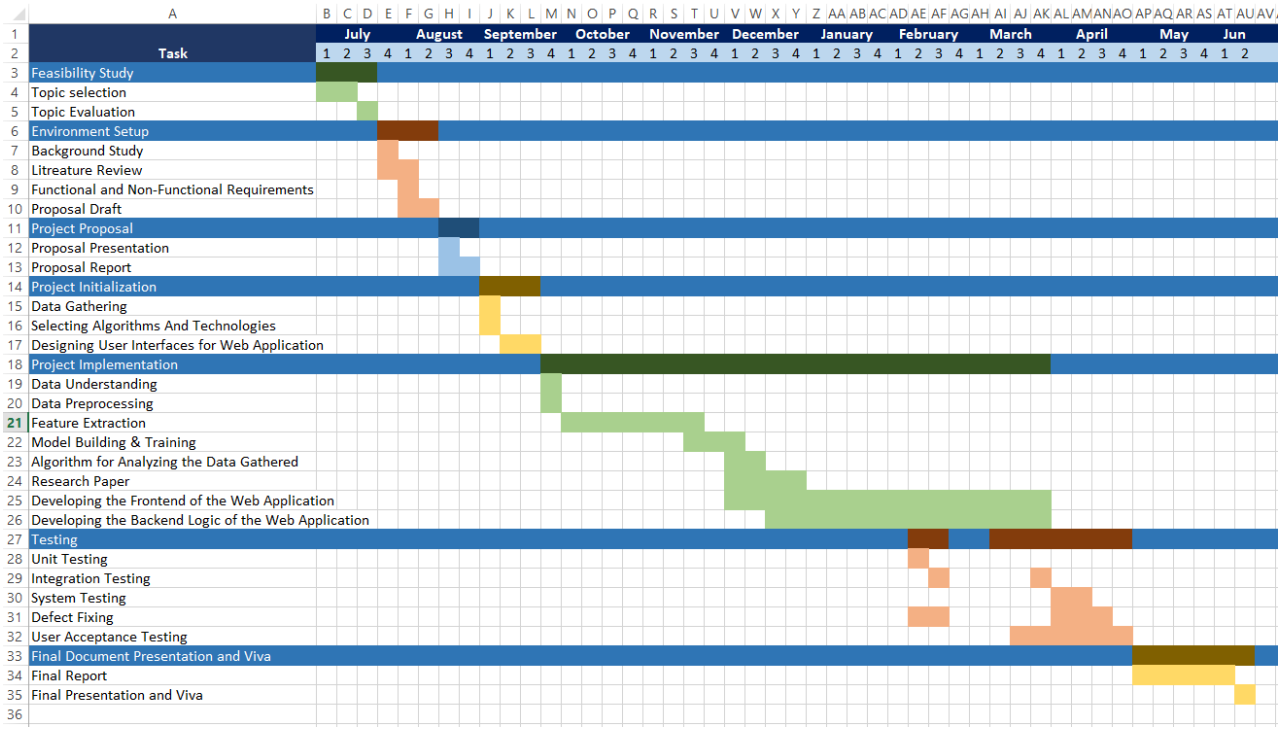

Figure 2 -- Gannt Chart

6. WORK BREAKDOWN CHART

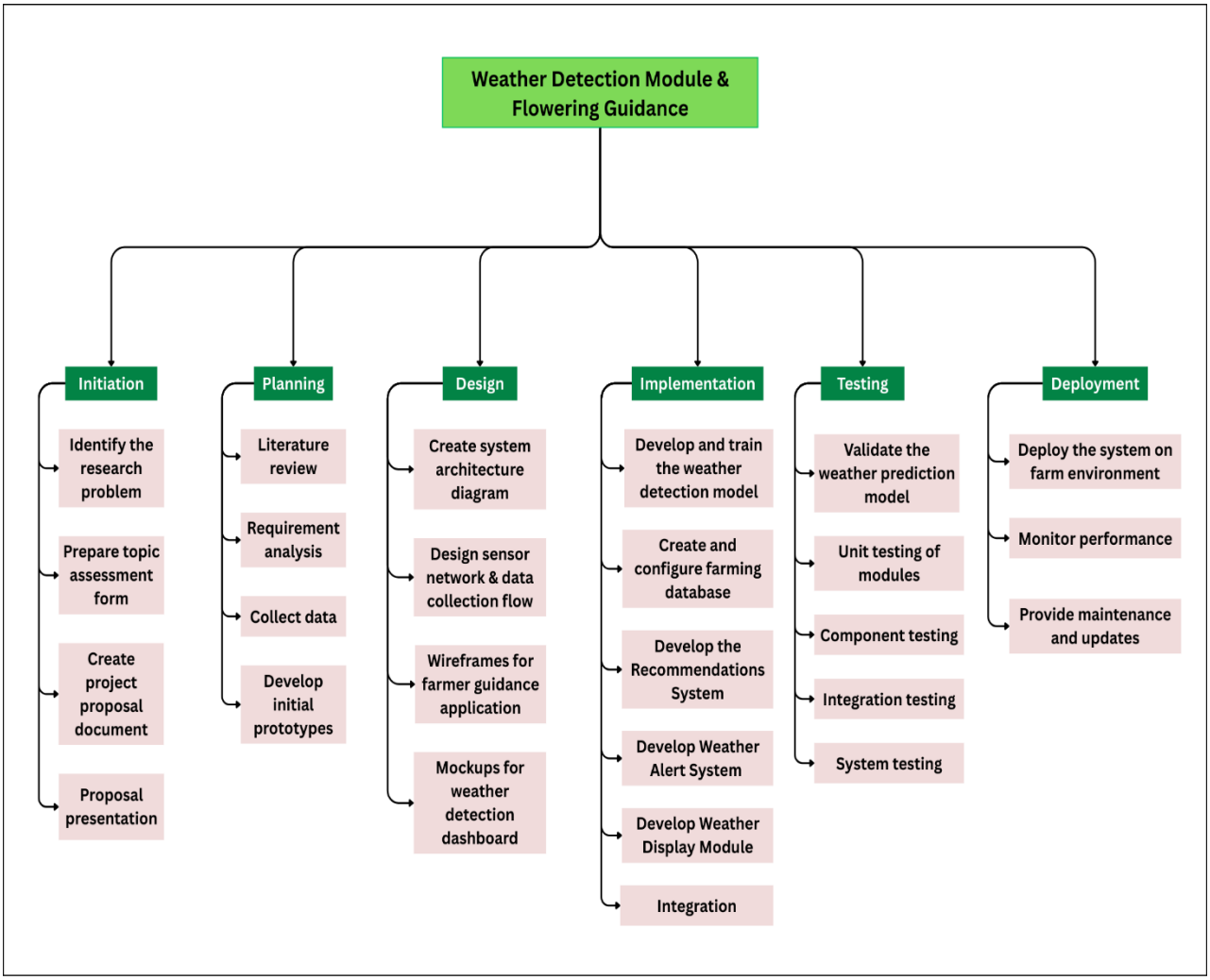

Figure 3 -- Work breakdown Chart

7. DESCRIPTION OF PERSONAL AND FACILITIES

Table 1 --- Associated tasks with the component

Sub-Component	Purpose	Associated tasks
Data Collection & Preprocessing Module	Gather, clean, and organize historical & real-time weather for the farm location(s).	Connect to weather APIs; set geolocation (lat/long) of greenhouse/shade-house; fetch current, hourly, and daily data; handle missing values, unit normalization (°C, %RH, mm, km/h), time zone alignment; cache responses and respect rate limits.
Weather API Integration Module	Provide reliable programmatic access to weather.	Create API keys and environment secrets; implement calls for current conditions, 48-hr hourly, 7–14-day daily forecasts, and alerts; add fallbacks/multi-provider strategy; error logging & retries. (Examples: OpenWeatherMap, Tomorrow.io, Meteostat; local Met Dept feed if available.)
Guidance & Recommendation Engine	Turn detections into clear actions for farmers.	Rule-based + simple ML (optional) to generate "do/avoid" guidance: adjust shade cloth or misting, irrigation timing, ventilation/fans, disease-risk sanitation, fertilization windows, harvest/transport timing; include why + data behind each tip.
Alerting & Notification Module	Deliver timely warnings and tips.	Set alert types (heat stress, chilling risk, very high RH, heavy rain, strong wind); severity levels; channels (SMS, WhatsApp, email, in-app); quiet hours; bilingual templates; escalation if conditions persist.
Visualization & Dashboard	Make data understandable at a glance.	Build mobile dashboard: current status, next 48 h timeline, 7-day outlook, VPD band, risk badges, actionable checklist; export PDF/CSV weekly report; role-based access for growers/extension officers.
System Testing & Maintenance	Ensure system reliability and freshness.	Unit tests for API parsers & calculators; synthetic data tests for rules; monitor uptime and API usage; rotate keys, update thresholds with agronomist feedback; quarterly review.

8. BUDGET AND BUDGET JUSTIFICATION

For this research component, we do not aim to commercialize our individual components and the whole research will be commercialized as an application. Following is the budget justification for the whole research system.

Table 2 --- Budget Table

Description	Cost
Cost for Deployment	10000.00
Internet	4000.00
Travelling cost for data collection	4000.00
Google Map API	2000.00
Paper Publishing	20000.00
Total	40000.00

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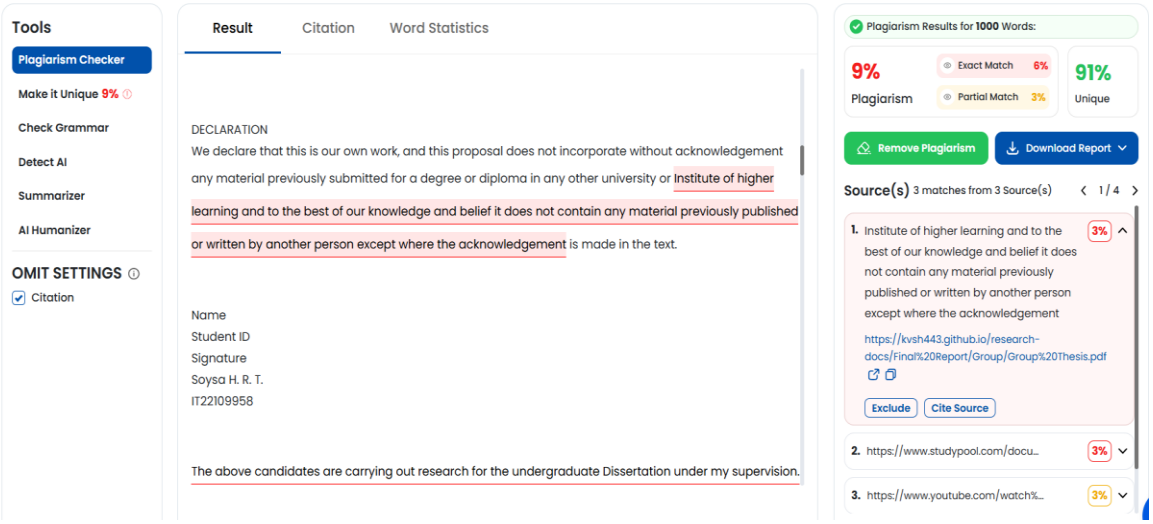
10. APPENDICES

Appendix - A: Plagiarism Report



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